Paper Review • arXiv 2611.06918 • 2026-11-04

From Pixels to Actions

Towards Native Vision-Language-Action Policies

NEO-VLA: No Vision Encoder; Actions Read Layer-wise Backbone Visual Computation

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Encoder-free 2B backbone | Layer-wise KV reading | Flow-matching expert

TL;DR

In short: actions read a native backbone's layer-wise vision

NEO-VLA drops the standalone vision encoder: light-conv patches evolve with language in a 2B, 40-layer backbone; a flow-matching action expert reads its KV per layer.

Finding: control-critical visual signals often peak in early layers, not the final representation; even frozen, native backbones beat encoder-based ones for action learning.

90.9

Frozen • LIBERO average

Same expert and data: Qwen3-VL-2B 79.5, Gemma-3-4B 84.5

94.9

Frozen, first-28 only

Beats uniform 28 layers (90.9) and last 28 layers only (86.7)

89.7

LIBERO-plus total

Best in table; camera shifts 94.5 (π0.5: 78.4)

69.4

RoboTwin 2.0 C2R

Best in table on randomized scenes; Clean 75.78, only ~6 pts drop when randomized

Motivation: mainstream VLAs pass only the encoder's final output

Mainstream approach

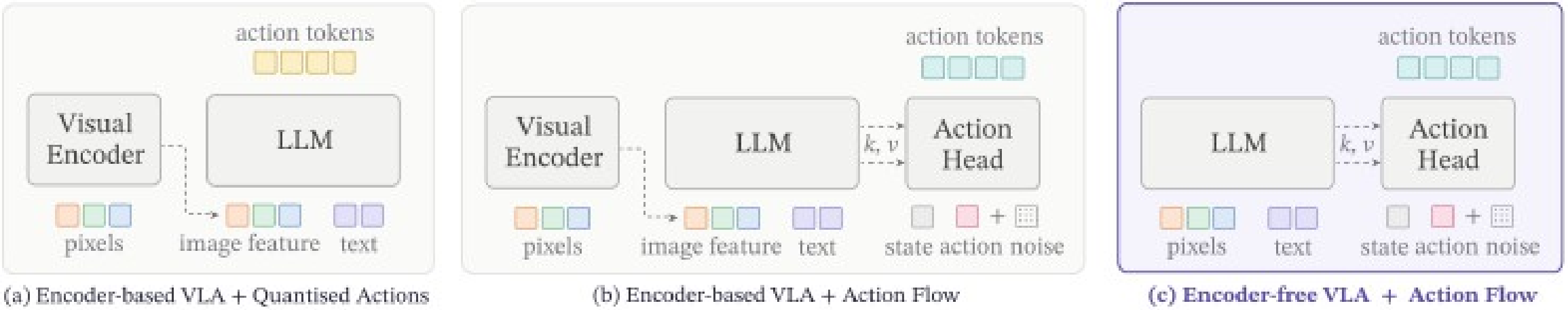
Image → CLIP / SigLIP / DINOv2 → LM backbone → action head.
Actions see only the encoder's final output; the visual hierarchy behind it is hidden.

Risk

Manipulation relies on local geometry, edges, depth and subtle state changes, which are easily abstracted away.
Evidence: vision is a bottleneck, pruning-sensitive, and harmed by action fine-tuning.

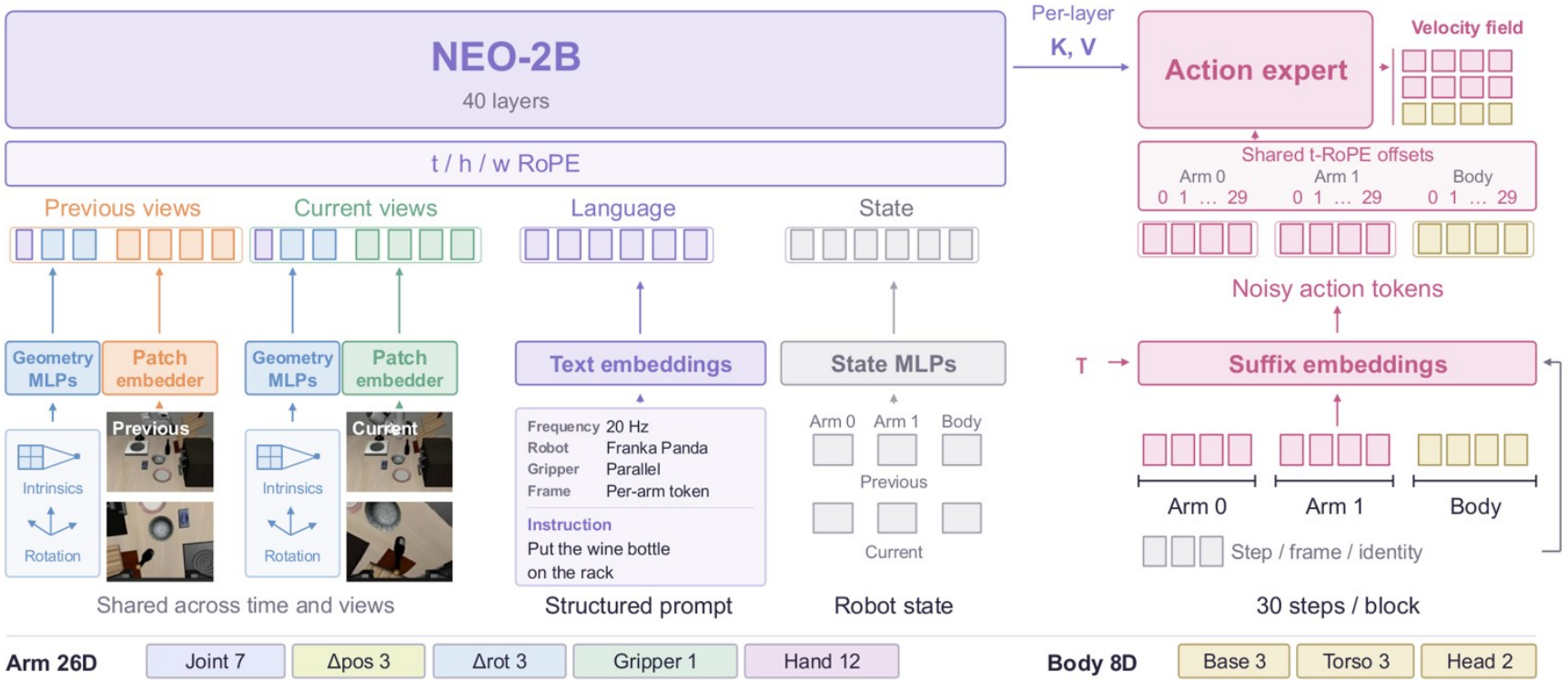
Core question

Should an action policy inherit a fixed visual endpoint,
or directly access the evolving visual computation?



(a) Encoder + discrete action tokens (b) Encoder + continuous action head (c) NEO-VLA: encoder-free; expert reads layer-wise KV, flow matching outputs continuous actions

NEO-VLA: pixels in, continuous actions out, no vision encoder



Native vision interface

Two-stage conv patch embedder, total stride 32; aspect ratio kept, ⩽88 tokens/view; no ViT / VAE / VQ codebook

Shared backbone

NEO-2B, 40 layers; t/h/w Native RoPE; causal text attention + bidirectional within image blocks

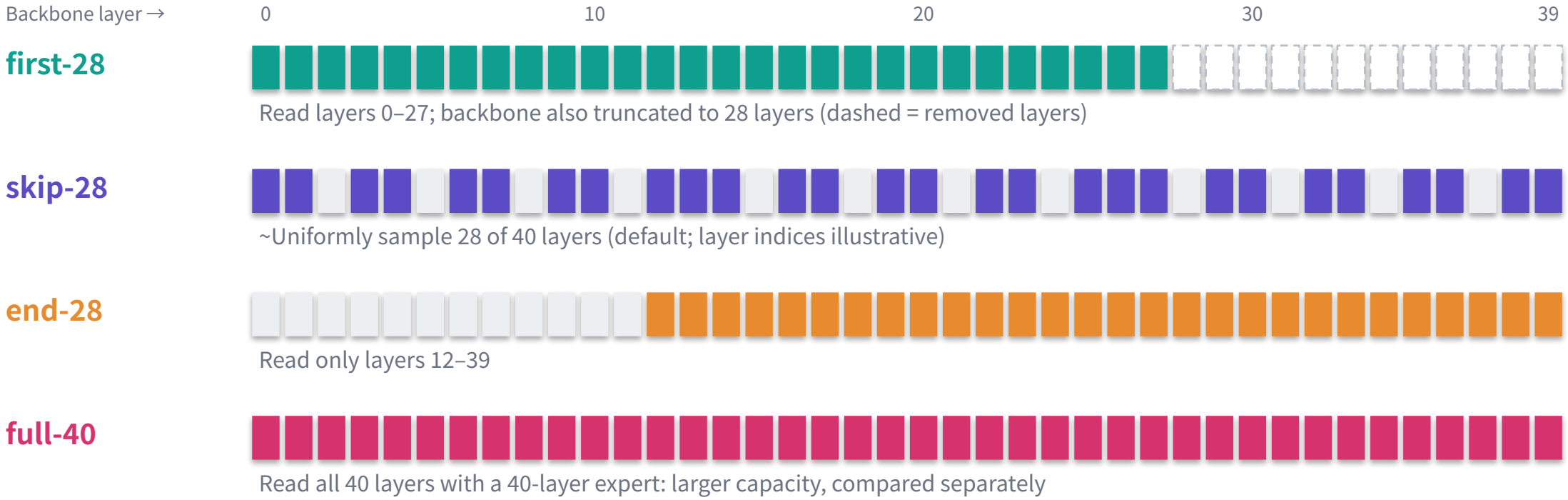
Unified multimodal prefix

Multi-view × 2 frames + cam. intrinsics/rotation MLP; prompt (rate/body/frame/instr.); 6 state tokens

Flow-matching expert

Reads backbone KV per layer; arms 2×30×26 + body 30×8; 10-step Euler inference, no CFG

Key design: which layers the action expert reads is configurable



Flow matching objective

$a_\tau = (1-\tau) \cdot \varepsilon + \tau \cdot a$, target velocity $v = a - \varepsilon$

$\tau = 0.999u$, $u \sim \text{Beta}(1, 1.5)$; four loss groups weighted (2, 1, 1, 1), weight 2 for end-effector & gripper

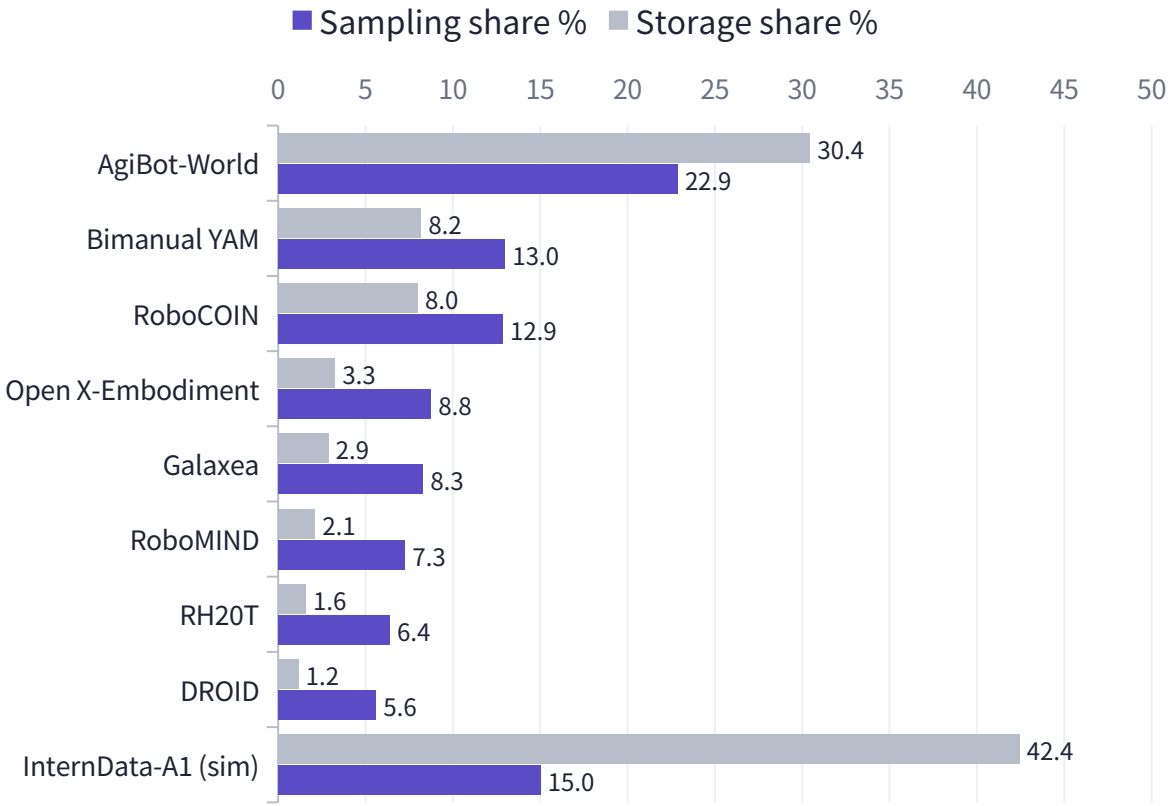
Unified action space (30-step chunks)

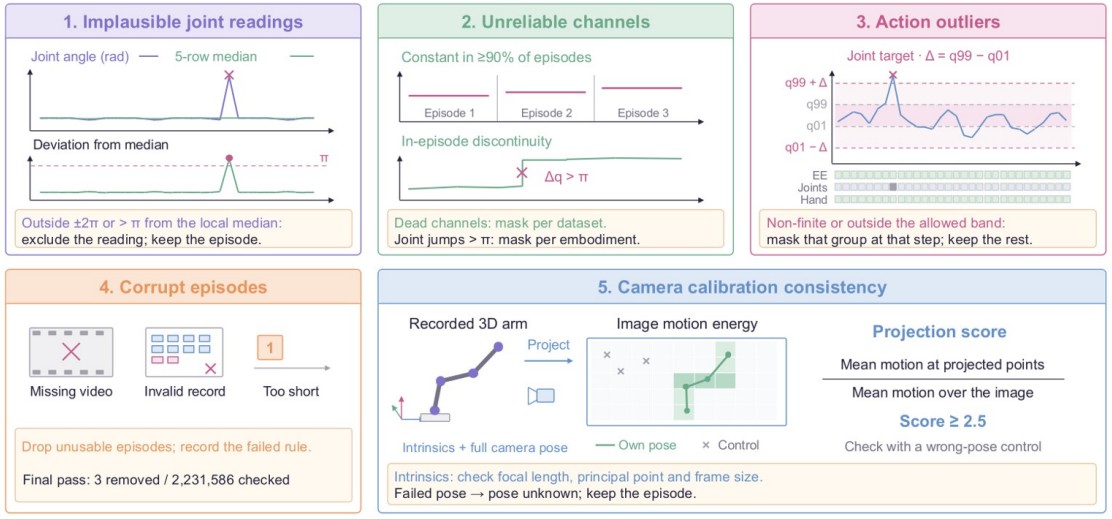
Per arm (26-D): 7 joints + 3 Δpos + 3 Δrot + 1 gripper + 12 hand joints

Body (8-D): 3 base + 3 torso + 2 head; absent parts masked per embodiment

Data: 9 public datasets, 2.23M trajectories, ~9,928 hours

2.23M	9,928 h	928M
episodes	Duration	Rows





Cleaning rule: mask only faulty parts, never drop whole episodes

- Joint reading beyond ±2π or >π from 5-row local median: drop it
- Channels constant in ⩾90% of episodes: mask per dataset; jumps >π: mask per embodiment
- Out-of-range or non-finite actions: mask only that group and timestep
- Camera poses scored by projected motion (⩾2.5 trusted; wrong poses as control); failures marked unknown
- **Final check: only 3 of 2,231,586 removed**

Training: expert first with frozen backbone, then full joint training

① Vision-language backbone (NEO-2B, 40 layers, encoder-free)

Large-scale image-text pretraining
Builds visual perception, keeps language ability

Mid-training
Captioning / dialogue / OCR / detection

Instruction tuning (SFT)
High-quality multimodal interaction

② Action training (robot corpus)

Stage 1 • Frozen backbone

- Train only expert & conditioning
- 113,400 steps • batch 8192
- Peak lr 2e-4

Stage 2 • Full + co-training

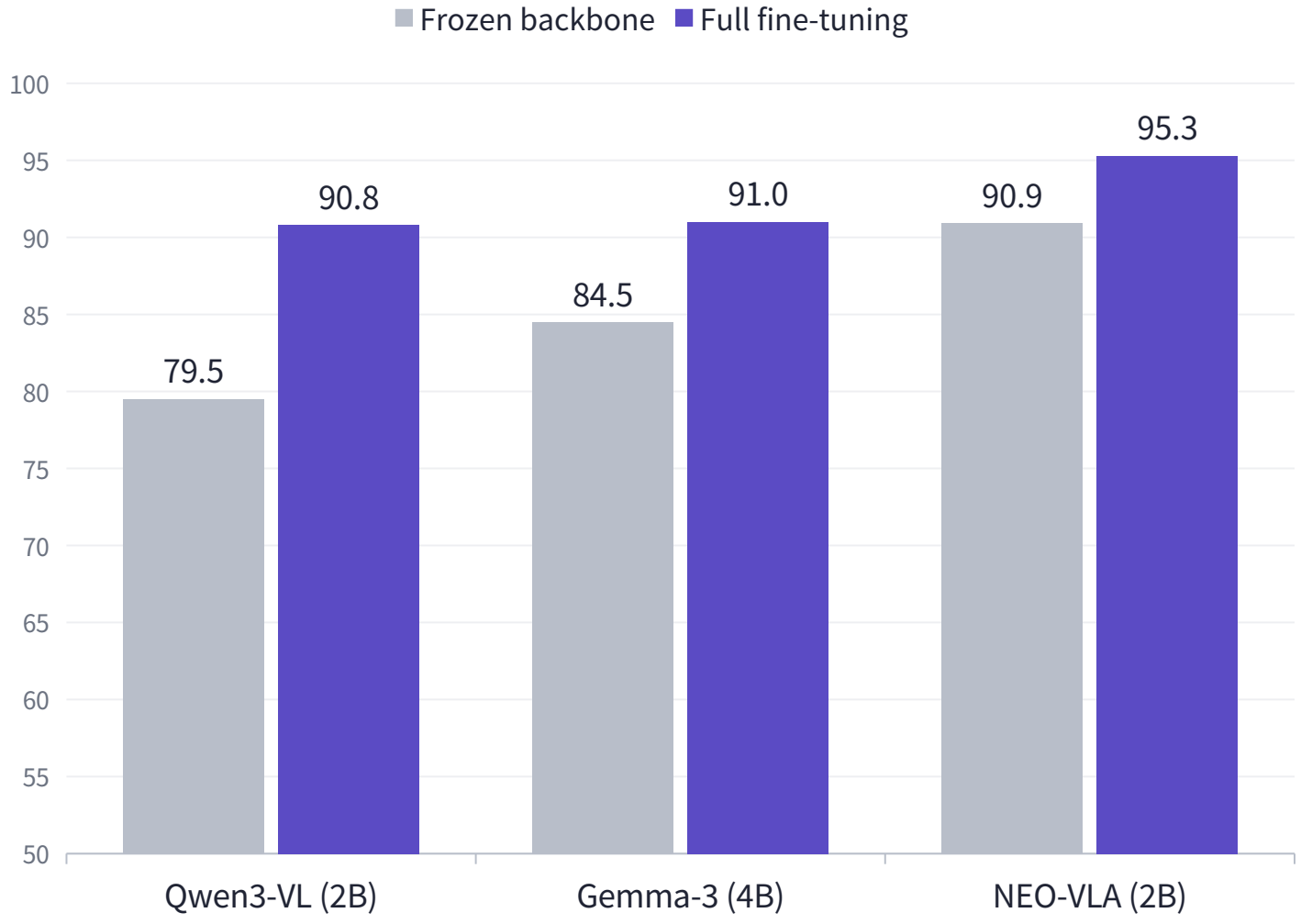
- Unfreeze all; same robot corpus
- 75,000 steps • batch 8192
- Expert lr 2e-4, backbone 5e-5
- VL data every other step, NTP ×0.1

Stage 3 • In-house data

- Init from Stage 2, full FT
- Switch to in-house action data
- 75,000 steps, else as Stage 2

AdamW (β1=0.9, β2=0.95), 2k-step warmup then cosine decay to 10% of peak, bf16 mixed precision, gradient clipping 1.0

Same setup: frozen native backbone ≈ fully tuned encoder VLMs



Frozen NEO ≈ fully tuned encoders

90.9 vs full FT 90.8 / 91.0

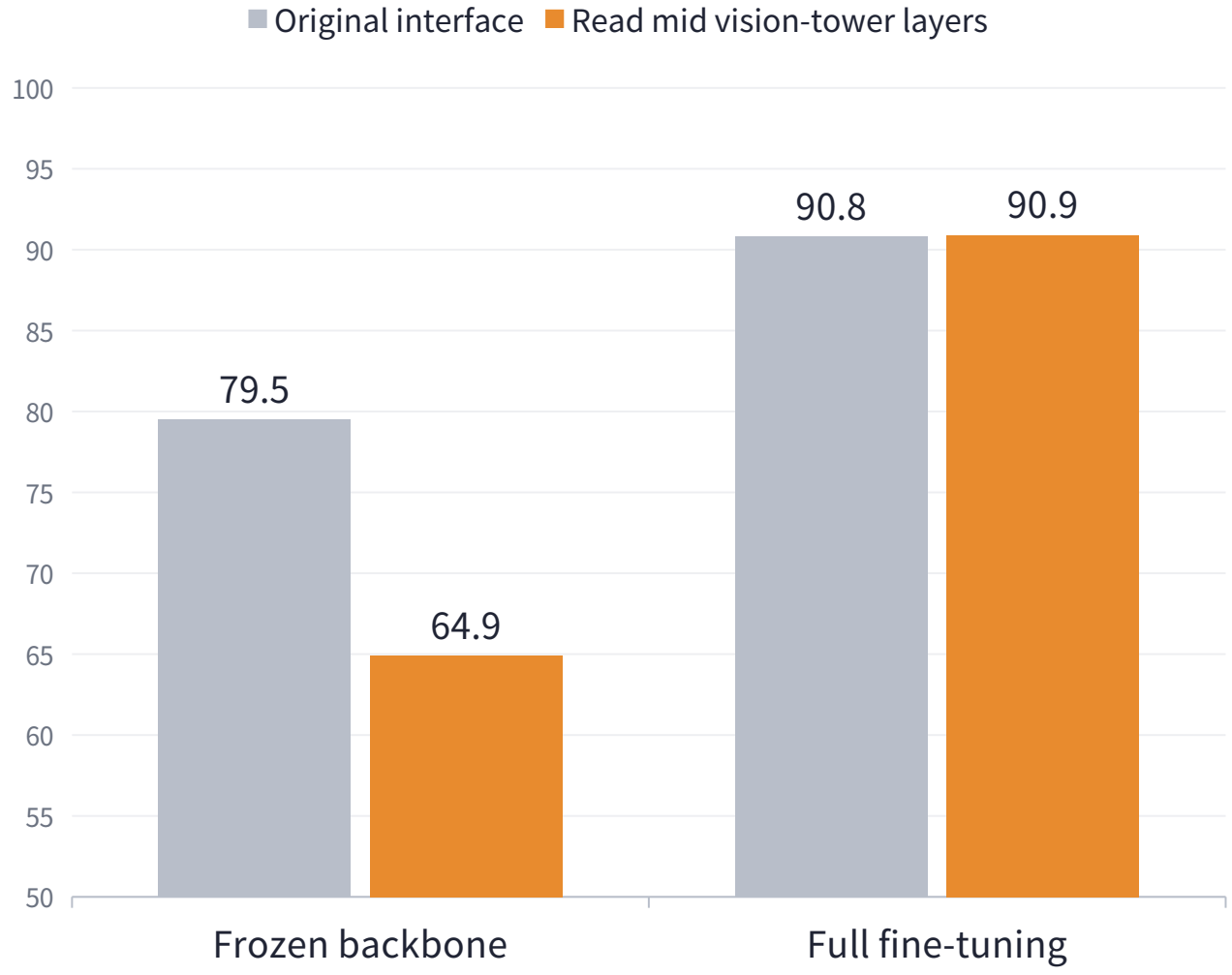
Largest gap on long-horizon

LIBERO-Long frozen 85.4 > Qwen / Gemma full FT 78.6 / 82.8

Beware confounders

The backbones differ in pretraining data and methods; authors admit gains can't all be credited to removing the encoder. Shared setup: 75k OXE action steps, 30k-step checkpoint, 2,000 episodes per config.

Counterexample: Qwen reading vision-tower layers is worse frozen



Setup

Qwen3-VL-2B's 13 expert layers read intermediate vision-tower features instead; no added parameters, all else equal.

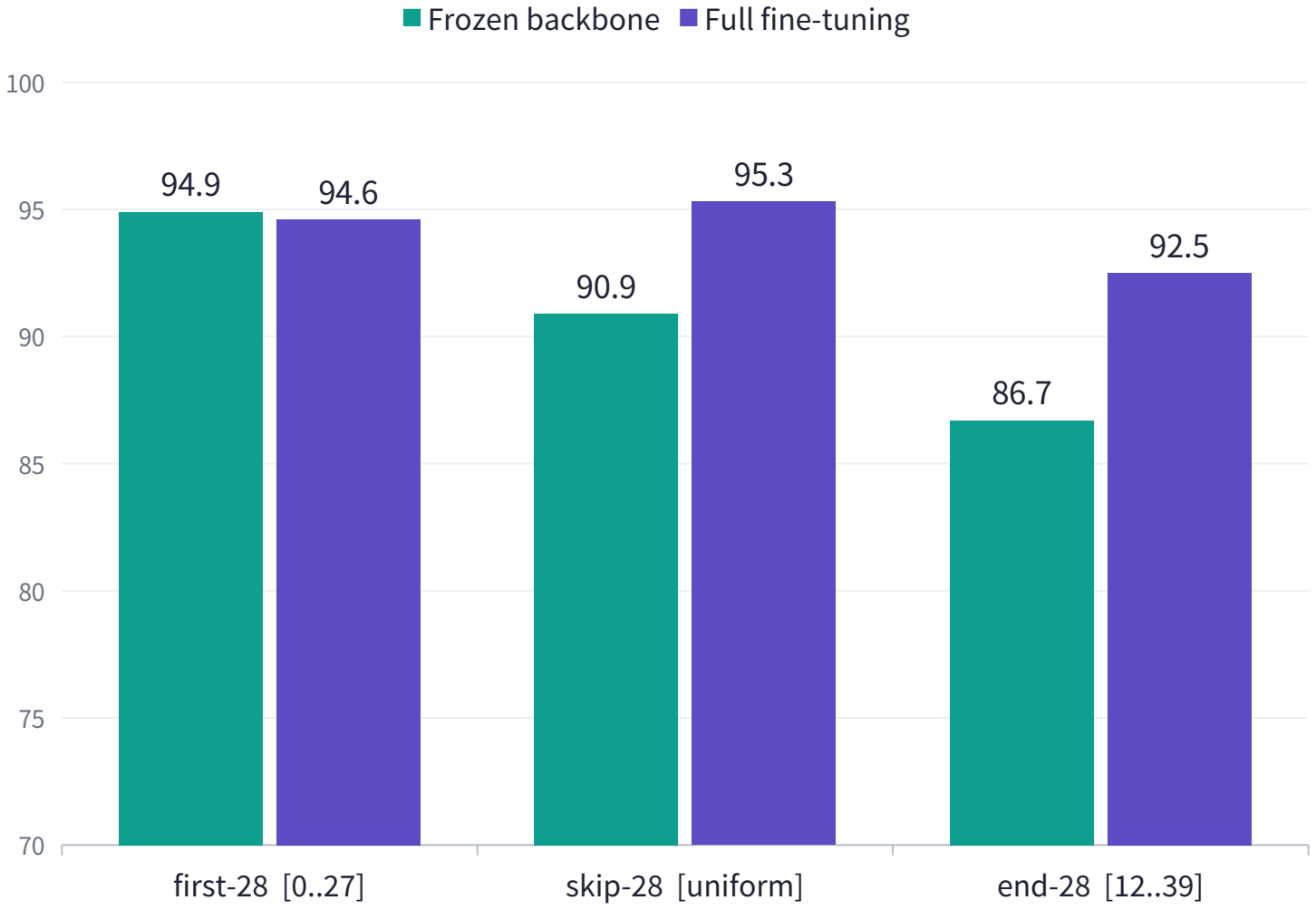
−14.6

Frozen: drops (79.5 → 64.9); full FT: on par (90.8 vs 90.9)

Paper's conclusion

NEO's edge is not just "access to early features": encoder-free pretraining shapes them into representations directly usable for control.

Read depth: early layers best frozen; early + late best fine-tuned



Non-overlapping control (frozen)

93.6
first-14 [0..13]

89.6
last-14 [26..39]

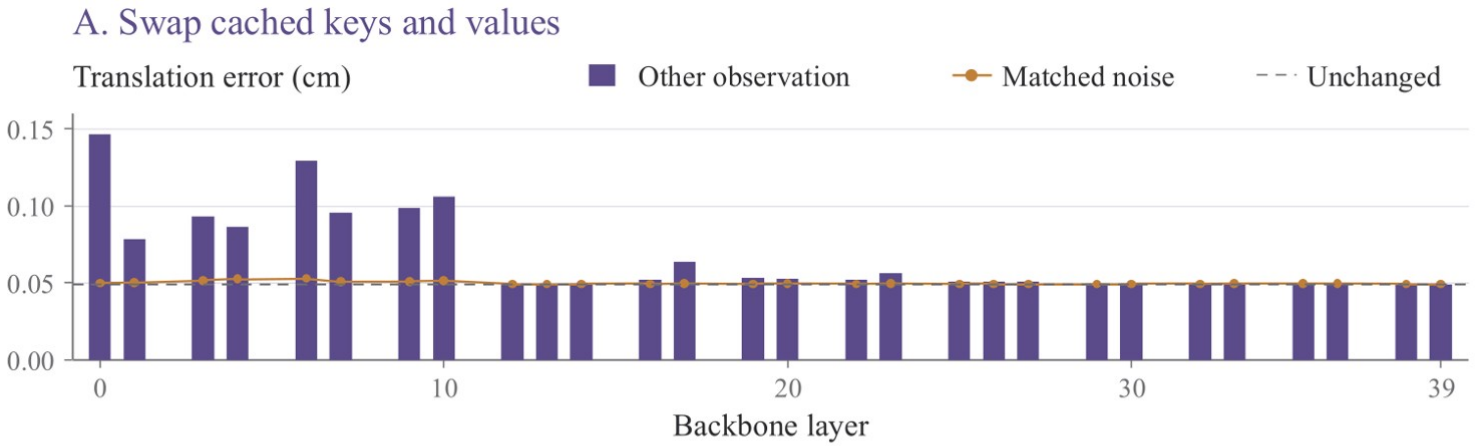
Conclusions differ by setting

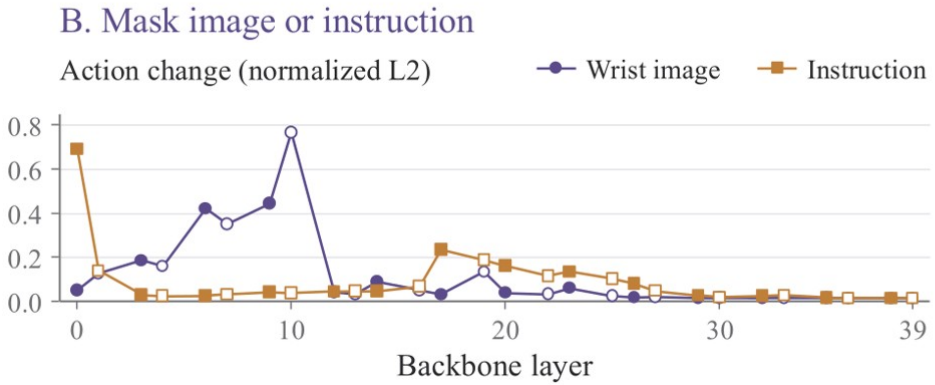
Frozen: early layers only is best; full FT: uniform full depth is best (95.3); end-28 is always worst.

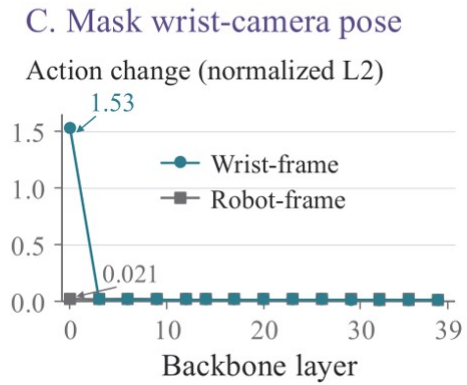
Confounder

first-28 / first-14 also truncate the backbone (depth 28 / 14), so read position and compute are not fully disentangled.

Interventions: swapping KV of the first 10 layers hurts most







A Swap cached KV

Swapping layers 0–10 raises translation error up to ~0.145 cm (baseline ≈0.05 cm); little effect after layer 11. Matched noise shifts ⩽0.004 cm, so content matters.

B Mask image / instruction

Instruction sensitivity peaks at layer 0 (≈0.69), minor peak near layer 17; wrist-image sensitivity at layers 6–10 (peak ≈0.77).

C Mask wrist camera pose

Wrist-frame model: 1.53 at layer 0; robot-frame model: only 0.021. Geometry is used in the earliest layers and depends on the task frame.

Action-relevant information spans depth, concentrated in early backbone layers

Preserving understanding underpins action robustness

69.4

Co-training • RoboTwin C2R

Understanding largely preserved

10.7

Action-only FT • RoboTwin C2R

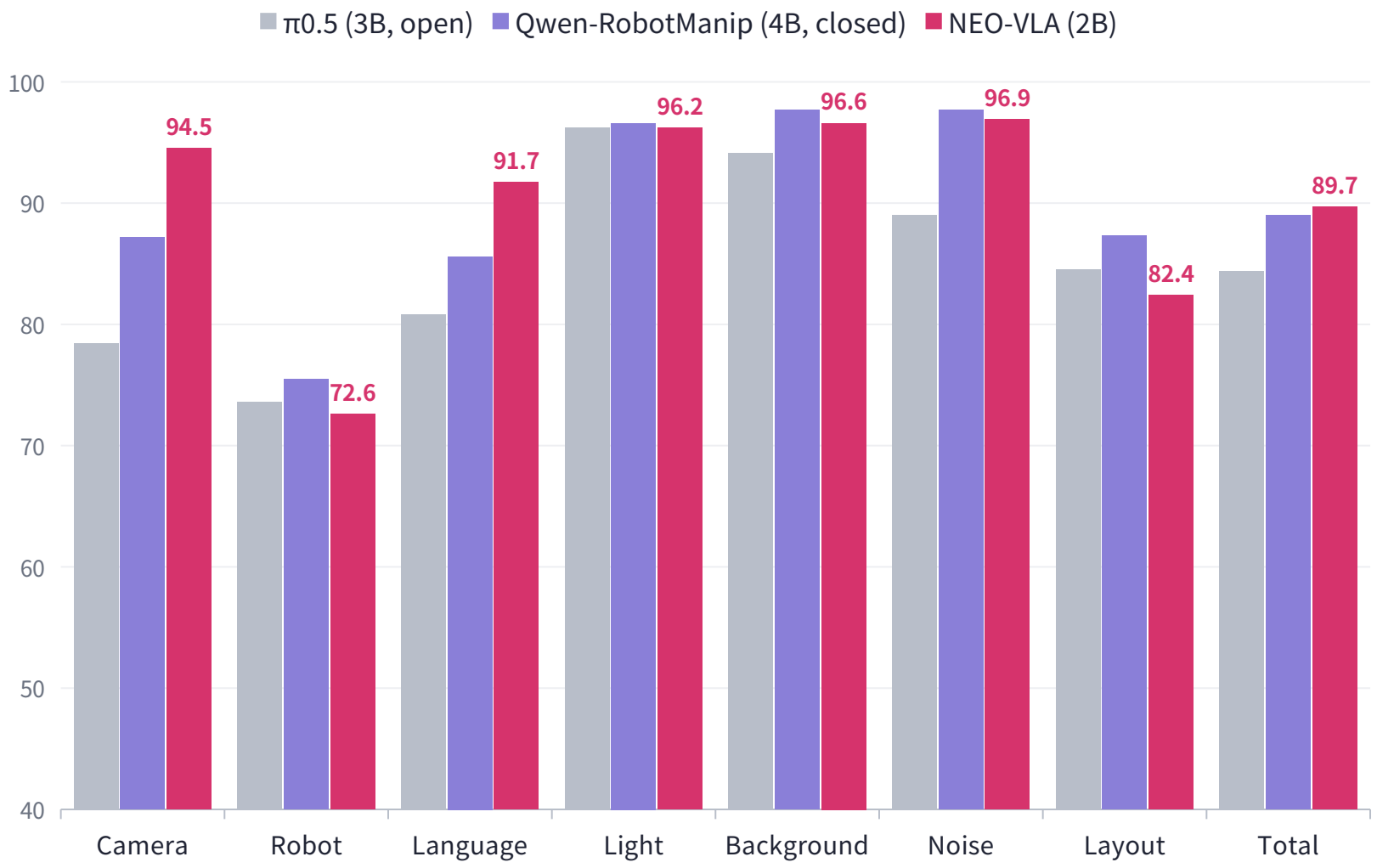
and model can no longer generate valid text

Multimodal benchmarks (%): co-training drops all within 1.5 pts

Benchmark	Before post-training	+ Co-training	Change
MMBench-EN	82.7	81.2	−1.5
RealWorldQA	66.8	68.4	**+1.6**
AI2D	82.9	82.3	−0.6
TextVQA	76.6	76.4	−0.2
ChartQA	80.8	80.4	−0.4

Paper's conclusion: multimodal understanding and robust action are not independent; preserving backbone understanding during post-training markedly improves policy generalization.

LIBERO-plus: largest gains under camera and language shifts



89.7

Total: best in table

Qwen-RobotManip 89.0 (closed)

+16.1

Camera 94.5 vs π0.5 78.4

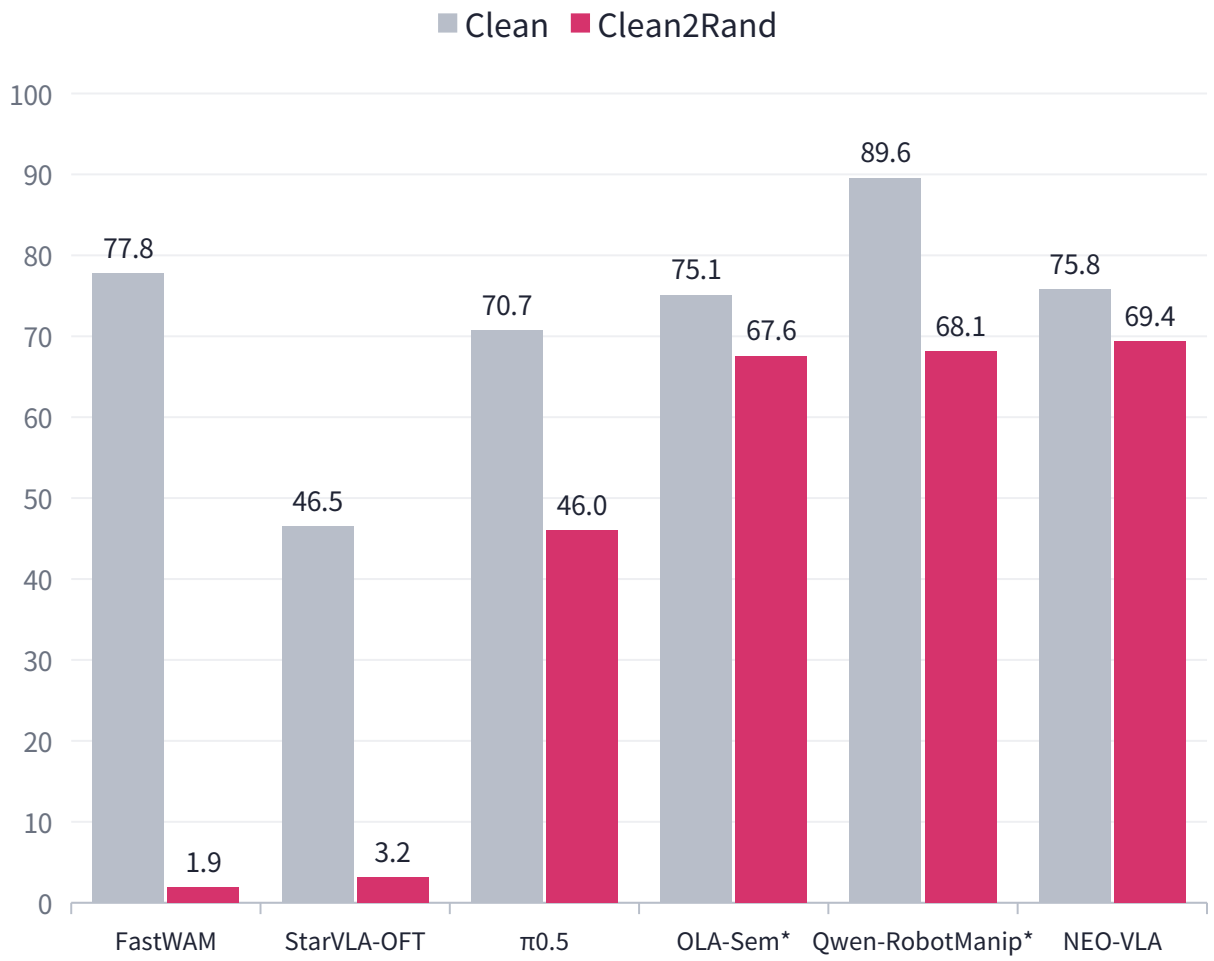
Language 91.7, also the best

Weaknesses

Robot 72.6 and Layout 82.4 are not the best: gains shrink when embodiment or layout changes.

Other benchmarks: robust to randomization, weak on position/task

RoboTwin 2.0: Clean → Clean2Rand (* closed-source)



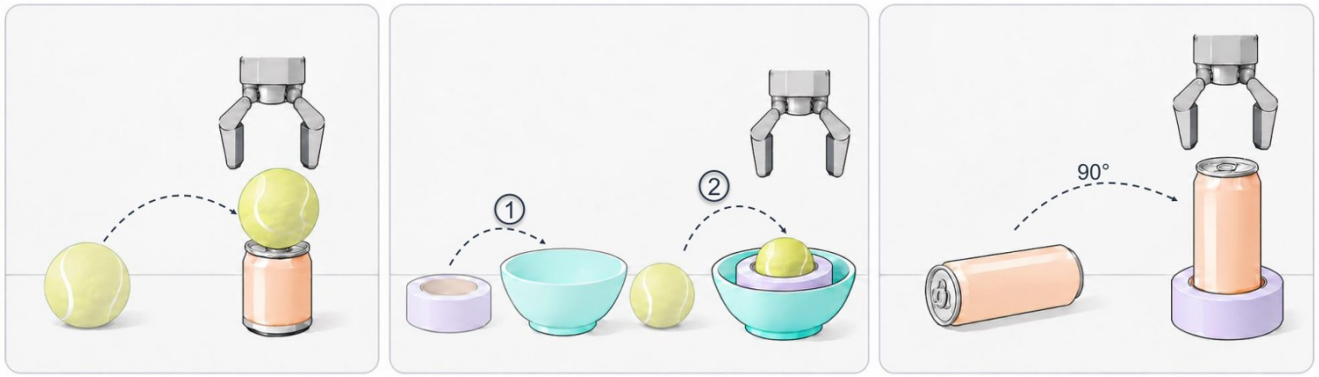
LIBERO: near saturation

Model	Size	Avg. success
OpenVLA	7B	76.5
π0.5	3B	96.8
MMaDA-VLA	8B	98.0
EO-1	3B	98.4
Qwen-RobotManip (closed)	4B	99.1
NEO-VLA	**2B**	**97.4**

LIBERO-Pro: fails on position / task shifts

Model	Object	Position	Semantic	Task	Avg
OpenVLA	93.0	0.0	97.3	0.0	47.6
π0.5	96.0	20.8	95.8	0.8	53.3
X-VLA	79.0	0.8	96.8	6.8	45.8
NEO-VLA	**84.3**	**3.0**	**98.0**	**6.3**	**47.9**

Real robot: 50 demos per task, matches or beats π0.5 on all 3



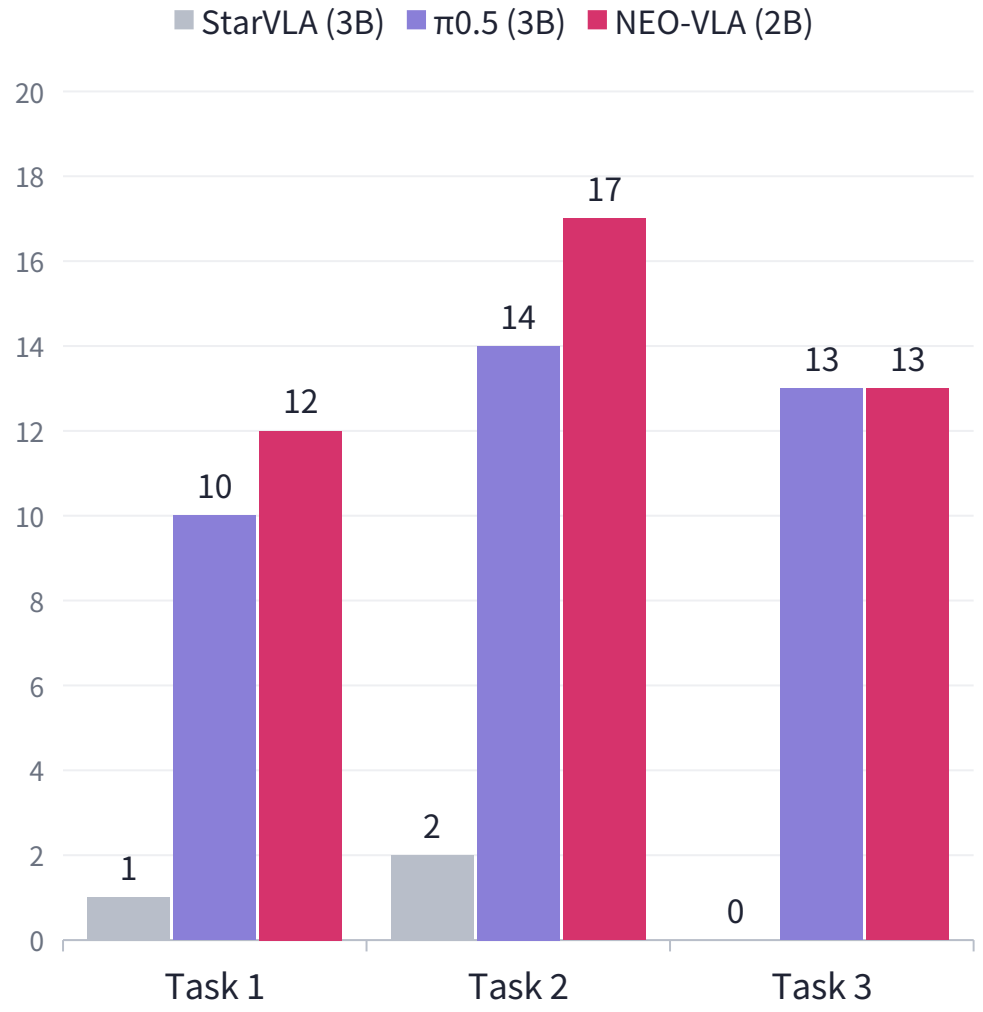
(a) Task 1: Tennis on can (b) Task 2: Tape in bowl, Tennis in tape (c) Task 3: Stand can in tape

Task 1 Place a tennis ball on a can (placement precision)

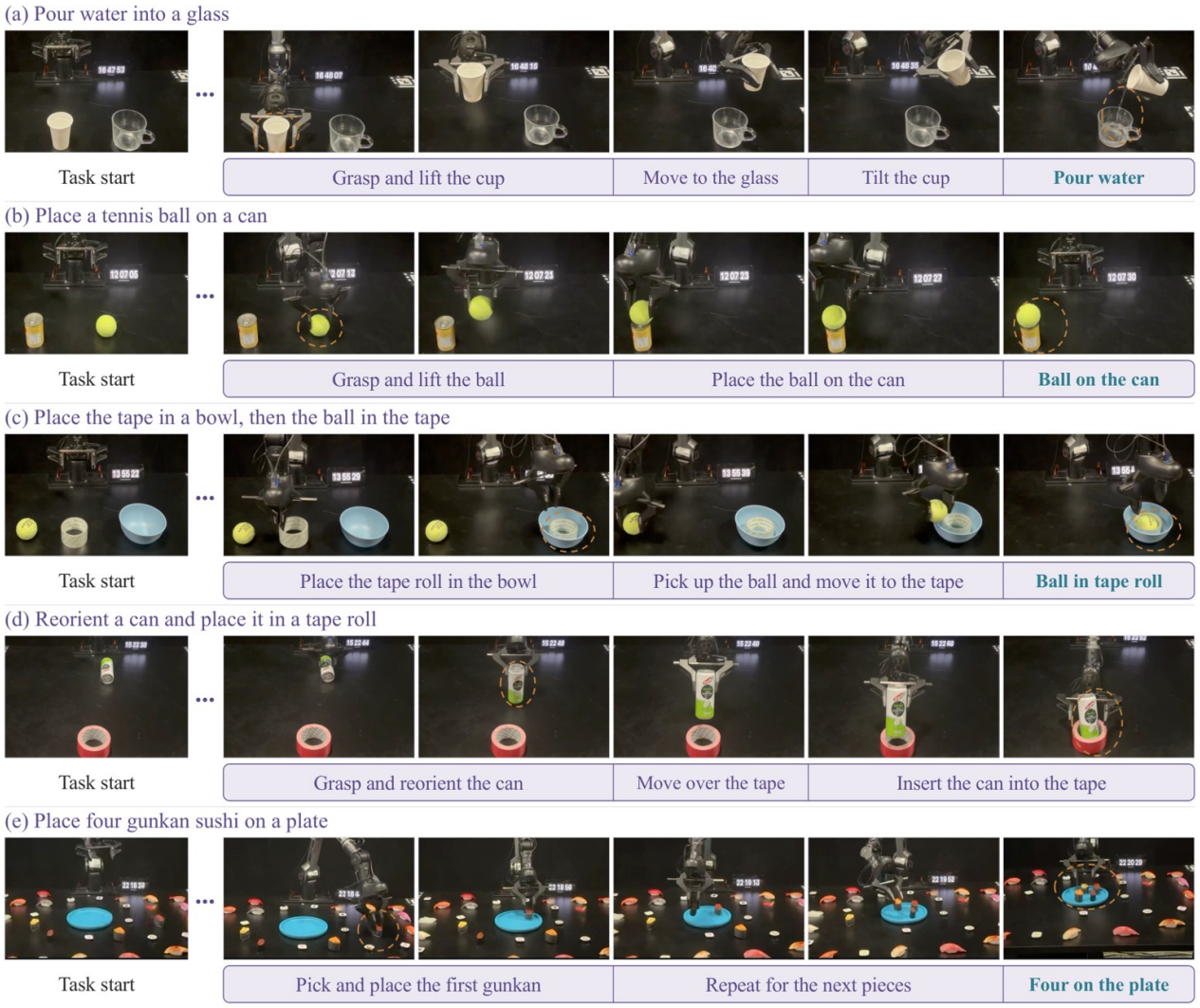
Task 2 Tape roll into bowl, then ball into tape roll (multi-step, long-horizon)

Task 3 Rotate a lying can 90° and insert it upright into a tape roll (reorientation)

Setup Fine-tuned on 50 demos per task; 20 trials = 10 in-domain + 10 out-of-domain; each method uses its recommended post-training recipe; 10-step Euler inference



Qualitative: pouring, placing, multi-step and reorientation



a Pick up a paper cup, pour into a glass

b Place a tennis ball on a can

c Tape roll into bowl, then ball into the roll

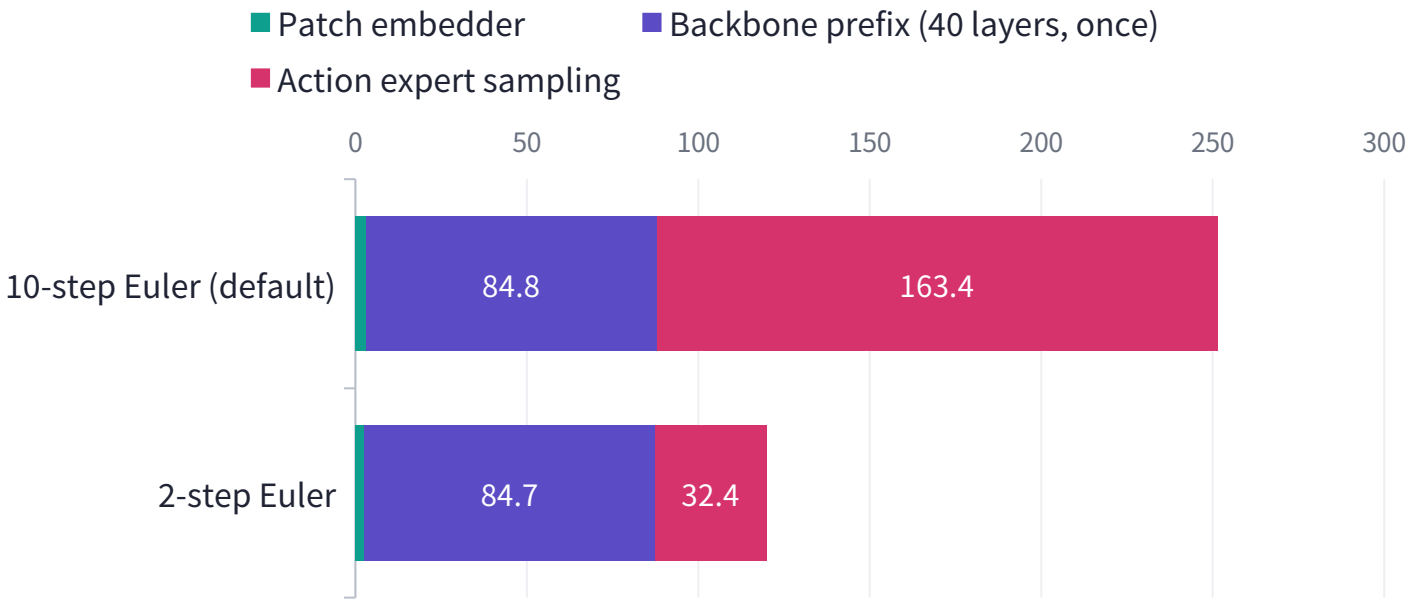
d Reorient a lying can, insert upright into tape roll

e Pick 4 gunkan sushi one by one from mixed sushi onto a plate

Note: (a) and (e) are qualitative only; no success rates reported.

Deployment cost

Inference latency: ~0.28 s per query on an RTX 3090



Patch embedder only 3.1 ms (10 steps) / 2.7 ms (2 steps); bars too thin to label

Observation: at 10 steps the action expert dominates cost; at 2 steps the backbone prefix dominates.

Trade-off: fewer steps change sampled actions; real-robot success (Table 10) uses 10 steps.

Note: stage timings are from Table 11; they do not sum to the full forward time.

End-to-end time (ms / query)

Stage	10 steps	2 steps
Forward (compiled)	282.5	147.8
Forward (eager)	515.4	210.7
Request → chunk	261.9	153.5
Request → exec. plan	**286.2**	**178.5**

≈0.29 s

Real robot: request → plan (10 steps)

Single RTX 3090, bf16, 30-step chunk per query, no real-time chunking

Our take

Verdict: a clear question, but attribution is still confounded

Strengths

1. The native architecture makes "which layer to read" a controllable variable, impossible for encoder-based VLAs
2. Layer-mapping ablation + KV swap / masking interventions + Qwen counterexample corroborate each other; matched-noise controls rule out "perturbation magnitude"
3. Robustness gains concentrate on camera / noise / language shifts, consistent with the mechanism analysis
4. Weights, code and pretraining data are all released; the unified action space and cleaning pipeline have engineering value

Caveats

1. Comparison with Qwen / Gemma is confounded by pretraining data and method differences (acknowledged by authors)
2. first-28 is best frozen but also truncates the backbone; interventions run only on the skip-28 model
3. LIBERO-Pro position 3.0, task 6.3: spatial and task-level generalization remain weak
4. Only 20 real trials per task; the drop to 10.7 with action-only FT is unexplained; ~0.28 s per inference

Three takeaways

1

Vision needed for control is spread across depth

and often peaks in early backbone layers; passing only the encoder's final output to the action head wastes it.

2

The key is not just removing the encoder

Vision and language must co-evolve in one backbone from the start; merely exposing middle layers of an encoder model is worse when frozen.

3

Next bottleneck: from perception to flexible action

NEO-VLA is robust to visual changes but clearly fails once object positions or task goals change.

Project page: https://www.lmms-lab.com/posts/neo_vla/ Code: https://github.com/EvolvingLMMs-Lab/NEO
Model: huggingface.co/weepiess2383/NEO-VLA Data: huggingface.co/datasets/weepiess2383/NEO-VLA-Action-Pretrain arXiv 2611.06918